



**MANIPAL  
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# **Classification of Astronomical Objects using Machine Learning**

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# Outline

- Introduction
- Literature Review
- Problem Statement
- Proposed Solution
- Objectives
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# Introduction

- Automates classification of stars, galaxies, and quasars using machine learning
- Utilizes labeled data from large-scale SDSS.
- Implements an end-to-end pipeline: preprocessing, model training, evaluation, and visualization
- Aims to improve accuracy and efficiency in astronomical data analysis
- Enables robust identification of celestial objects in expansive datasets

- Compares three feed-forward ANN variants: baseline, regularized, and GA-optimized
- All models trained on identical feature-engineered photometric dataset (100,000 samples, 13 features)
- Stratified 60-20-20 train/validation/test split ensures fair evaluation
- Genetic Algorithm used to automatically discover optimal ANN architecture
- Performance assessed using accuracy, precision, recall, F1-score, and confusion matrices
- Objective: identify ANN design choices that improve astronomical classification performance

# Literature Review

- **Literature Review - Decision Tree Approaches**

- Ball et al. (SDSS DR3 Classification)**

- **Objective:**

Classification of 142+ million astronomical objects from SDSS DR3 using machine learning

- **Methodology:**

Algorithm: Axis-parallel decision trees

Training data: 477,068 spectroscopically classified objects

Classes: Galaxies, Stars, QSOs, and NSNGs

- **Key Results:**

- **Completeness:** Galaxies (98.9%), Stars (91.6%), NSNGs (88.5%)

- **Efficiency:** Galaxies (98.3%), Stars (93.5%), NSNGs (94.5%)

Reliable classifications for magnitude  $r \leq 20$

Provided probability estimates for each class

- **Limitations & Insights:**

Inevitable extrapolation from spectroscopic to photometric regime

Performance varies significantly across different object classes

Required extensive supercomputing resources for parameter optimization



## Literature Review - Neural Network Approaches

### Cavuoti et al. (MLPQNA Classification)

#### Objective:

- Disentangle different spectral classes using neural network-based MLPQNA model

#### Methodology:

- Algorithm: Multi-Layer Perceptron with Quasi Newton Algorithm
- Input features: 10 photometric colors from optical data
- Three experimental categories for classification tasks

#### Key Results:

- Overall efficiency: 90%** across classification tasks
- Successfully identified 3.6M QSO candidates (529K robust candidates)
- 10 colors performed better than standard 4-color approach

#### Key Challenges:

- Unable to separate AGNs from normal galaxies using optical data alone
- Multiplicative propagation of errors in hierarchical classification
- Quality filtering required based on user-specific needs

#### Important Finding:

- Direct classification preferred over hierarchical pairwise approaches

# Problem Statement

The vast volume of astronomical data from modern sky surveys makes manual classification of stars, galaxies, and quasars inefficient and error-prone. This project aims to automate and enhance the accuracy of astronomical object classification using machine learning techniques.

# Objectives

- Develop and implement machine learning models to accurately classify stars, galaxies, and quasars from survey data.
- Create a robust data preprocessing pipeline including cleaning, normalization, and feature selection.
- Evaluate model performance using metrics like accuracy, precision, and recall to ensure reliable classification.
- Compare different classifiers including Decision Tree, Random Forest and ANN to identify the best-performing model.
- Identify ANN design choices that improve astronomical classification performance. Baseline, regularized, and GA-optimized.



# Proposed Methodology

## **Data Collection:**

Obtained labeled SDSS photometric data of Stars, Galaxies, and Quasars.

## **Preprocessing & Feature Engineering:**

Handled missing values, computed color indices and flux ratios, encoded labels, and applied RobustScaler normalization.

## **Data Splitting:**

Performed stratified 60-20-20 split into training, validation, and test sets.

## **Regularized ANN Design:**

Built deep feed-forward ANN with batch normalization, dropout, and L2 regularization to improve generalization.

## **GA-Based Architecture Optimization:**

Used Genetic Algorithm to evolve ANN architecture (layers, neurons, dropout, L2) based on validation accuracy.

## **Model Training:**

Trained ANN variants using Adam optimizer with EarlyStopping and model checkpointing.

## **Model Evaluation:**

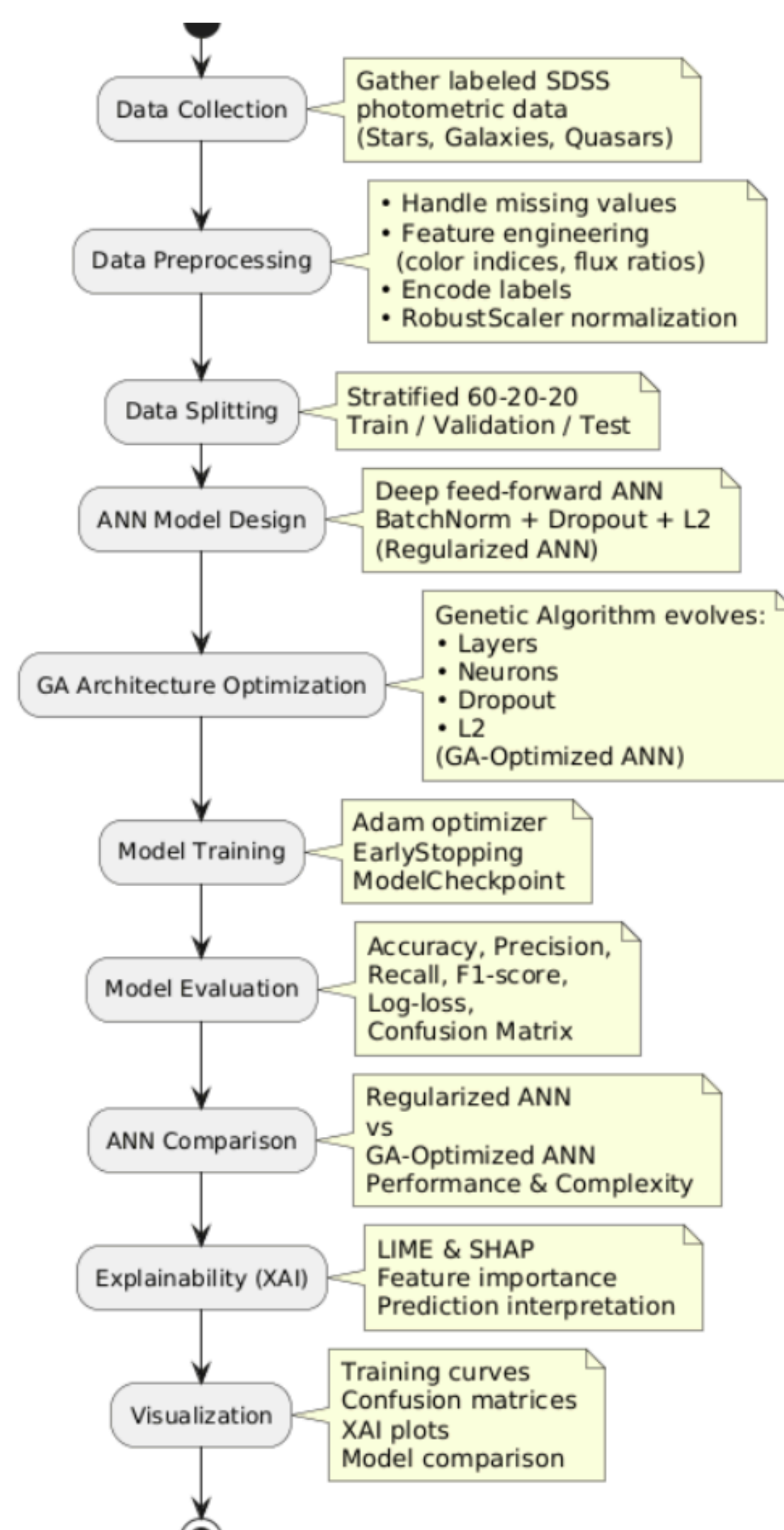
Assessed performance using accuracy, precision, recall, F1-score, log-loss, and confusion matrices.

## **Explainable AI Analysis:**

Applied LIME and SHAP to interpret feature importance and model predictions.

## **Comparative Analysis:**

Compared regularized ANN and GA-optimized ANN in terms of performance, overfitting, and complexity.



# Result

| MODEL                   | Training | Test  |
|-------------------------|----------|-------|
| Base ANN                | 97.19    | 97.42 |
| ANN with regularization | 97.58    | 97.61 |
| ANN with GA             | 97.16    | 97.21 |

# Outcome

## Research Paper:

A research paper on classifying stars, galaxies, and quasars using regularized and GA-optimized neural networks has been drafted and is planned for submission next semester.

## Work Completed This Semester:

- Designed and trained a regularized deep ANN for astronomical data classification
- Developed a Genetic Algorithm to automatically optimize ANN architecture
- Applied LIME and SHAP to understand how the models make predictions
- Performed detailed comparison between ANN variants

## Planned Next Semester:

- Submit research paper to a suitable conference or journal
- Extend experiments on larger astronomical datasets
- Refine model evaluation and analysis

# References

1. Ball, N. M., & Brunner, R. J. (2006). Data Mining and Machine Learning in Astronomy. *MNRAS*, 372(2), 614-626.
2. Cavuoti, S., Brescia, M., & Longo, G. (2015). Machine Learning for SDSS Classification. *MNRAS*, 450(4), 3893-3905.
3. Du Buisson, S., et al. (2016). Automated Classification of SDSS Transients. *MNRAS*, 460(3), 3076-3092.



Thank You